Actuarial Statistics: Confidence Intervals in Health Insurance

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- Math 476

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Research Question

- Do standard confidence intervals capture expected medical claims per person?
- Test classical CI methods on realistic, skewed claim distributions
- Check whether they achieve intended 95% coverage
- Assess actuarial consequences when coverage fails

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Actuarial Context: Health Insurance

Claims

- Amounts billed for care
- Range from minor visits to major surgery

Premiums

- Monthly member payments
- Too low: insurer loses money
- Too high: customers leave

Actuary's Goal

- Estimate expected cost per member
- Keep pricing solvent and competitive

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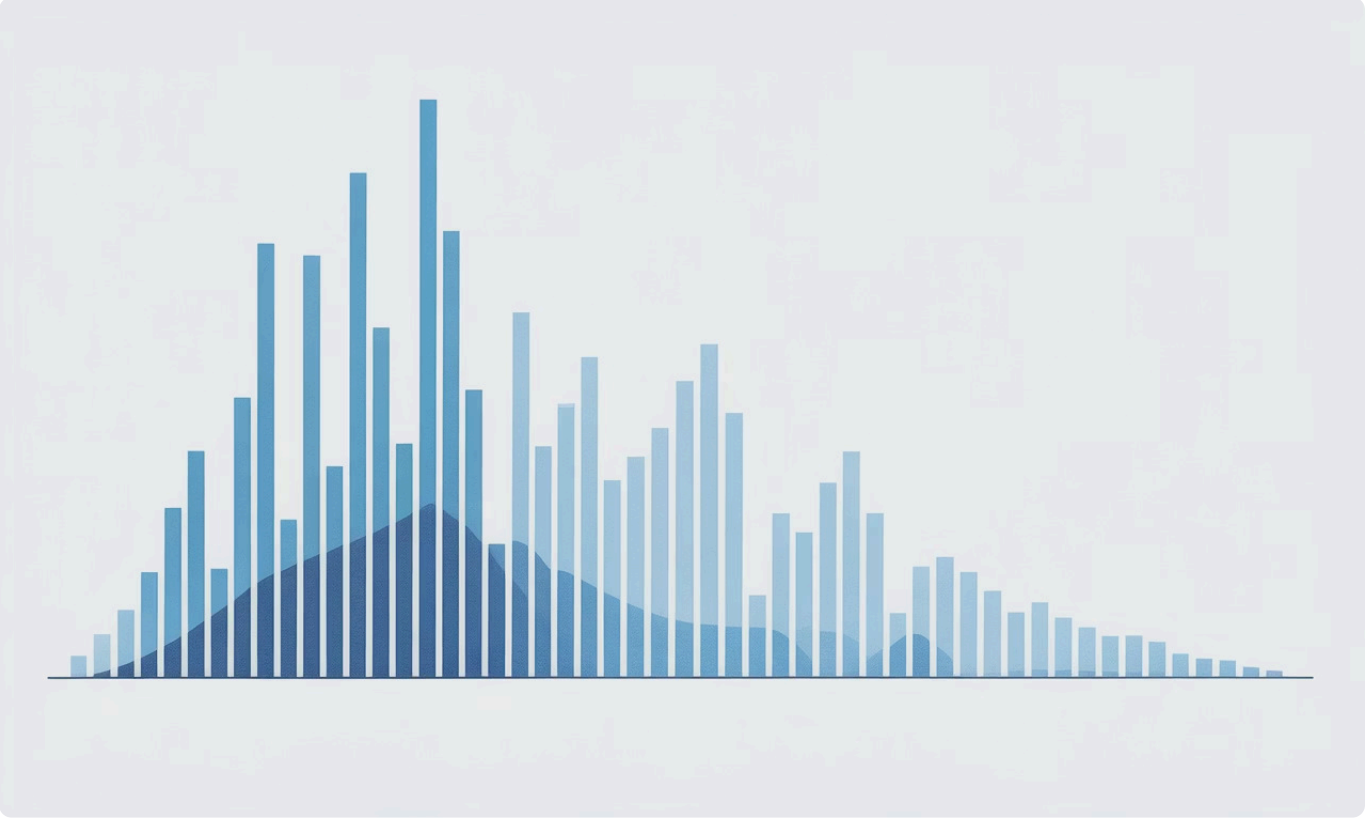
How Medical Claims Behave



Claim Severity

- Most claims are small
- Routine visits and prescriptions
- Few claims are extremely large
- Surgeries, cancer care, long hospital stays

The Result

- Claims are right-skewed
- Long tail of rare catastrophic events
- Outliers drive total cost

- Normal distribution does not model claims well

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Why Normal Fails — and What Actuaries Use Instead

- Normal distribution misses the long right tail of medical claims
- Actuarial models use heavy-tailed alternatives
- **Gamma** — mild skew; small-cost accumulation
- **Lognormal** — moderate to high skew; multiplicative growth
- **Weibull** — flexible; used for time-to-event and severity
- **Pareto** — heaviest tail; extreme outliers and catastrophic claims
- Focus: **Gamma** and **Lognormal**

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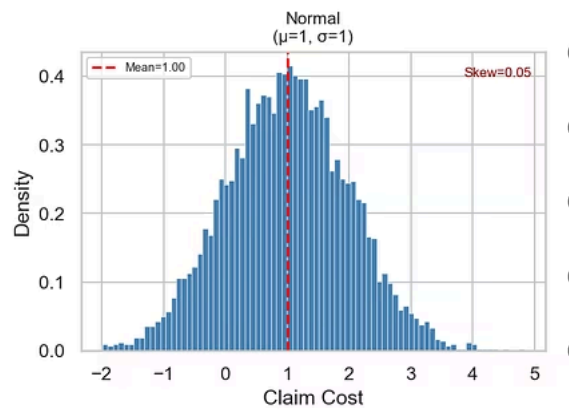
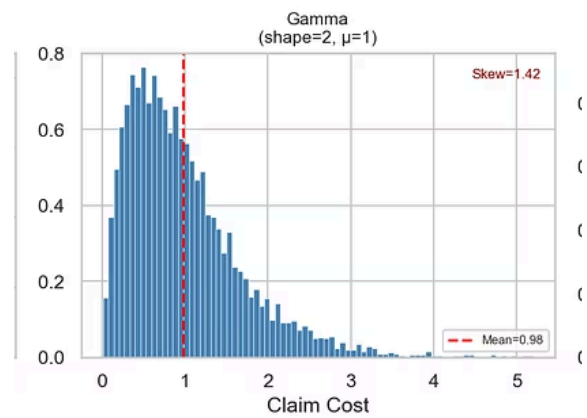
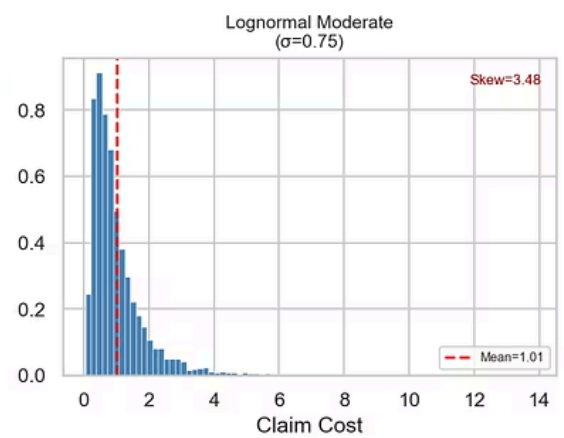
The Five Distributions We Model

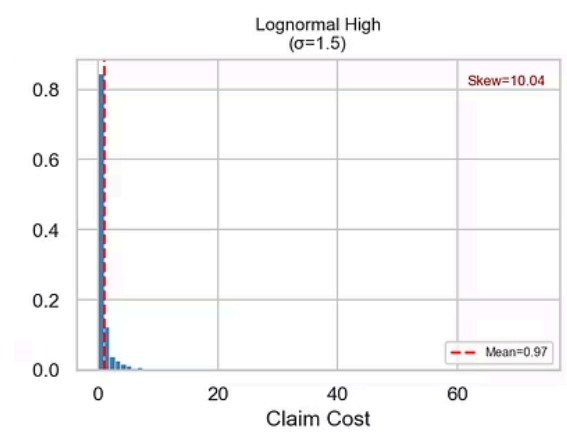
All distributions are scaled so the **true mean = 1**.

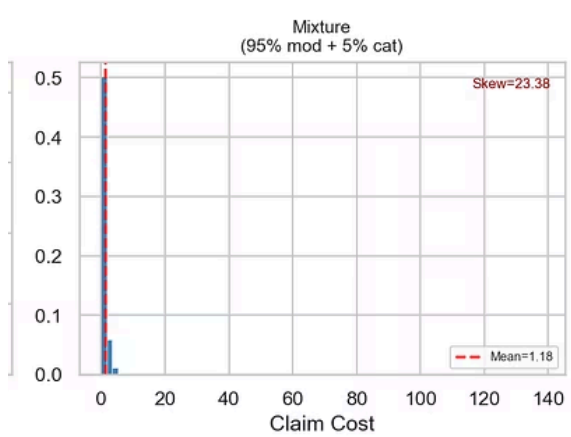
1	**Normal (μ=1, σ^2=1)** • Control case • Used to verify CI mechanics
2	**Gamma (shape=2)** • Mild skew (1.41) • Pharmacy claims
3	**Lognormal (σ=0.75)** • Moderate skew (2.5) • Common commercial population
4	**Lognormal (σ=1.5)** • High skew (11) • High-risk groups
5	**Mixture: 95% LN(μ=0.8, σ=0.75) + 5% LN(μ=4, σ=2)** • Most realistic • Top 5% drive ~50% of costs











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Why CI Accuracy Matters in Actuarial Practice

Underestimate Risk

- 80% CI instead of 95%
- Prices too low
- Insufficient reserves
- Possible legal liability

Overestimate Risk

- Premiums too high
- Customers leave
- Market share declines

The Stakes

- Highly regulated industry
- Mispricing affects solvency
- May trigger regulatory action

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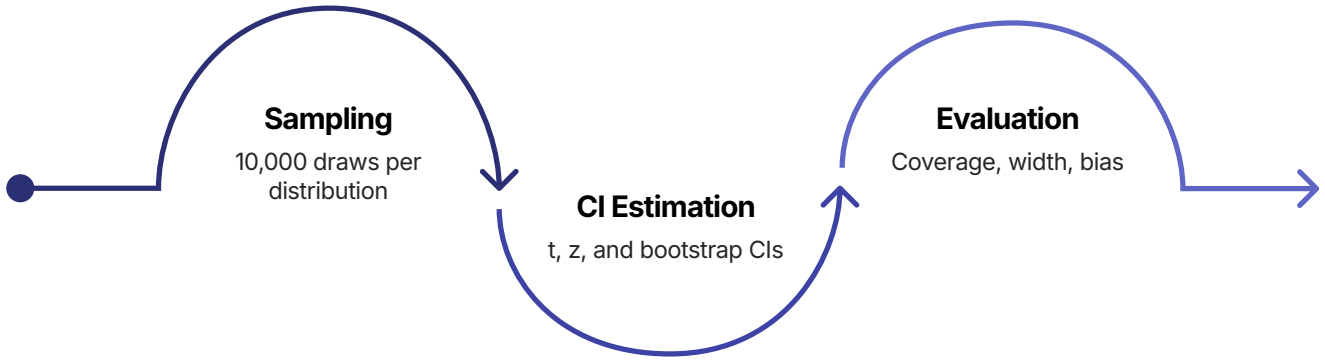
Simulation Design: Monte Carlo Framework



Sample Sizes Tested

- n = 25, 50, 100, 250, 500
- Small to large insurer groups

Key Metrics

- **Coverage**: CI contains true mean
- **Width**: Upper – lower bound
- **Bias**: Sample mean – true mean
- **Type I Error**: 1 – coverage

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Three CI Methods Compared

1

Z-Interval

- Fixed critical value: 1.96
- Best for large samples
- Formula: $\bar{x} \pm 1.96 \cdot (s/\sqrt{n})$

2

T-Interval

- Critical value depends on n
- Wider at small samples
- Usually better coverage

3

Bootstrap

- Resamples data 499 times
- Uses 2.5th and 97.5th percentiles
- No distributional assumption

- 10,000 simulations per distribution and sample size
- All three methods run on each simulation
- Used to estimate coverage

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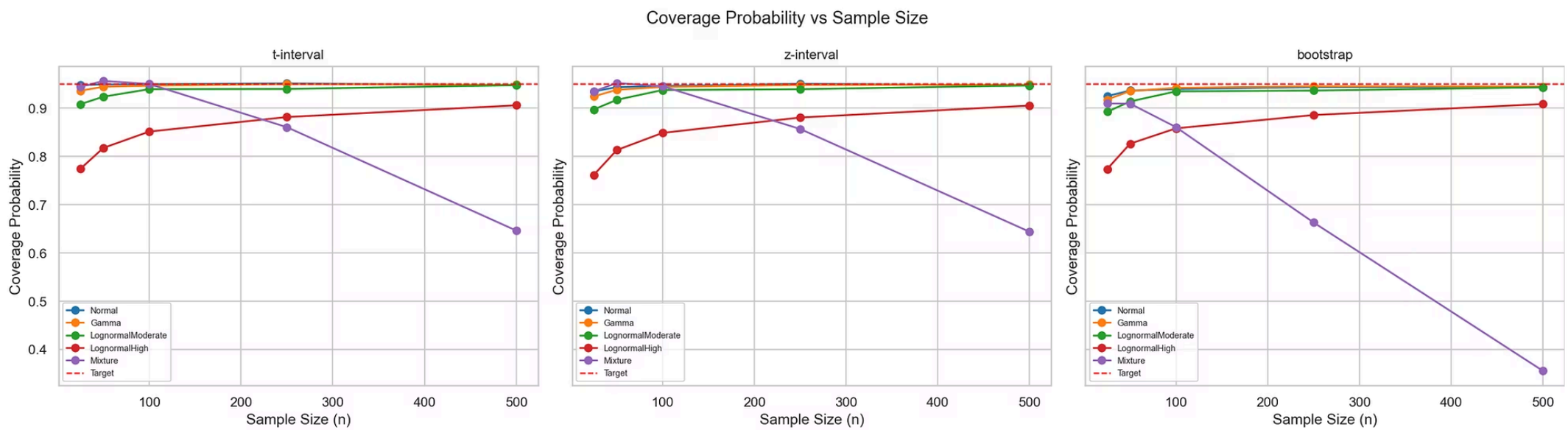
Main Result: Coverage



What We Found

- **Normal**: ~95% quickly
- **Gamma**: ~93% at n=25; ~95% by n=500
- **LN (σ=0.75)**: ~90% at n=25; ~94.5% by n=500
- **LN (σ=1.5)**: ~77% at n=25; ~90% by n=500
- **Mixture**: Falls to ~60% as n grows

- CLT helps unbiased estimators, as n grows, distributions converge to a normal approximation
- Mixture is structurally biased

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Bias: The Mixture Problem

Singular Distributions (Normal, Gamma, Lognormal)

- Unbiased estimators
- Sample mean → true mean ($\mu = 1$)
- Bias ≈ 0 at all sample sizes
- Mathematically guaranteed

Mixture Distribution

- Structurally biased
- Sample mean overestimates by ~0.25
- Bias persists as n grows
- Rare catastrophic claims pull the mean upward

- Mixture bias comes from structure, not sample size
- High variability does not remove the upward bias
- Sample mean centers above the true mean of 1

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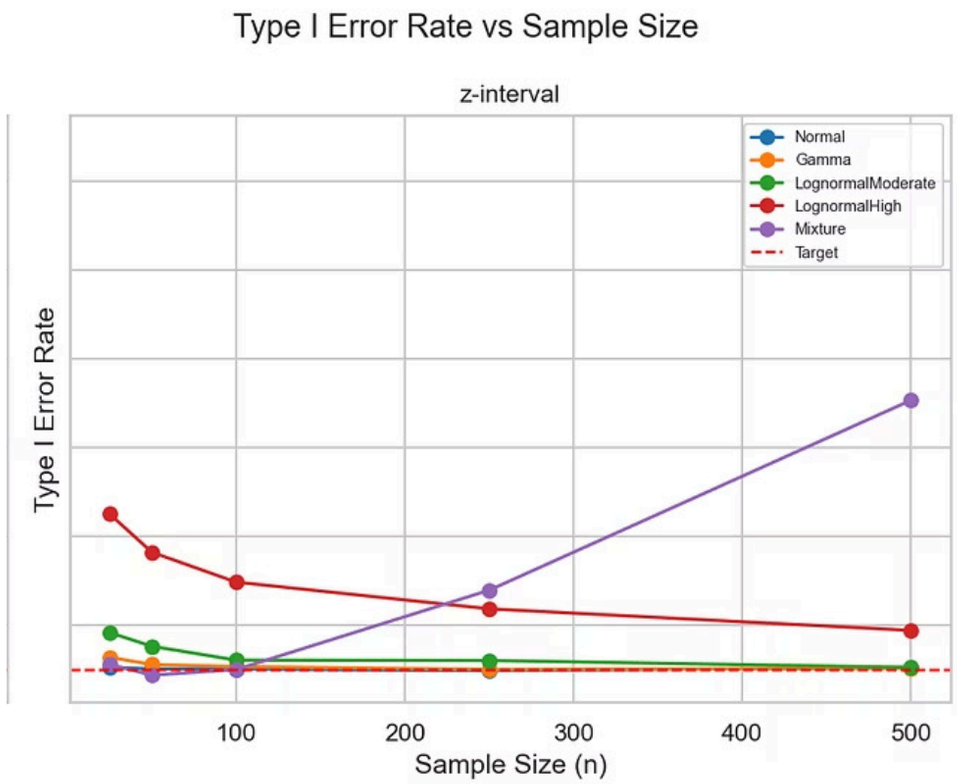
Type I Error: The Hypothesis Testing Connection



- **Type I Error = 1 – Coverage**
- Rejecting the true mean μ = 1

Confidence Intervals

- 95% CI should cover μ 95% of the time
- 85% coverage = 15% misses
- 15% Type I error instead of 5%

Hypothesis Testing

- Null hypothesis: μ = 1
- Missing μ = 1 means reject H_0 at α = 0.05
- Lower coverage = higher Type I error
- LN(σ=1.5), n=25: 77% coverage → 23% Type I error

- **Key insight:** CI failure and false rejection are the same event
- Broken CIs lead to invalid hypothesis tests

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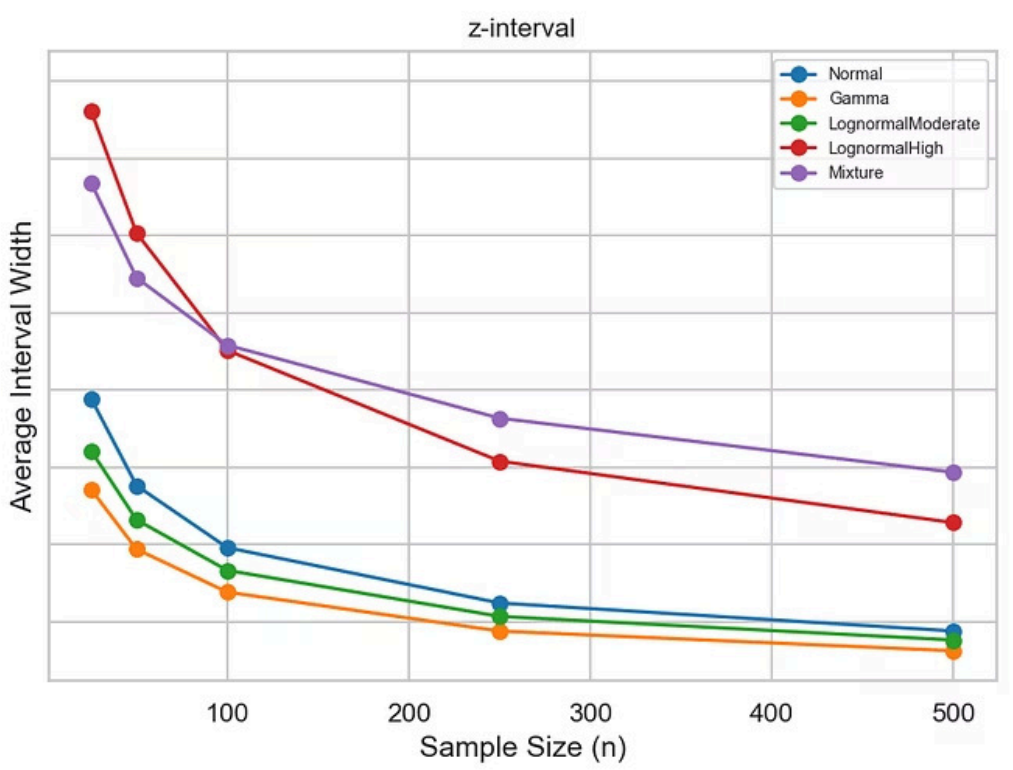
How Skewness Affects Confidence Interval Width

- **Key finding:** More skewness means wider confidence intervals



Normal Distribution

- Narrow intervals
- Width ≈ 0.85 at n=25
- Symmetric

Lognormal (σ=0.75)

- Wider intervals
- Width ≈ 1.2 at n=25
- Moderate skew

Mixture Distribution

- Widest intervals
- Width ≈ 1.55 at n=25
- Extreme skew

- Wider intervals do not always mean better coverage
- Mixture is widest at small n
- As n grows, intervals narrow
- Skewness increases variability
- Width alone does not guarantee coverage

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What These Results Mean for Actuaries

Finding 1 - Gamma Distribution

- Small samples (n=25): not reliable
- Larger samples (n≥100): works well
- Claims can be estimated with enough data

Finding 2 - Lognormal (σ=0.75)

- Common in commercial health insurance
- n=500 coverage: ~94.5%
- Only slightly below 95%

Finding 3 - Lognormal (σ=1.5)

- High-risk groups show severe undercoverage
- n=25 coverage: 77%
- n=500 coverage: 90%
- Standard CIs are unreliable

Finding 4 - Mixture Distribution (Most Realistic)

- Most realistic insurance portfolio model
- Standard CIs underestimate risk
- Small samples can look deceptively okay
- Coverage drops to ~60% as sample size grows
- Not reliable without adjustment

- Higher skewness = less reliable confidence intervals
- Mixture distribution is the biggest concern
- Standard CI methods should be adjusted or avoided

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Actuarial Implications: Why This Matters

Underestimating Risk

- 80% coverage instead of 95%
- Premiums too low
- Reserves too small
- Higher insolvency risk

Pricing Accuracy

- Premiums depend on expected cost per member
- Unreliable CIs make pricing unreliable
- Prices may be too low
- Or too high to offset risk

The Takeaway:

- Standard CIs are insufficient for mixture claims
- Use larger samples when possible
- Adjust methods for skewness
- Consider alternative estimation techniques
- Apply risk adjustments to premiums

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